

Chap. 5

Temperature Sensors

1. Introduction

2. Temperature Sensors

A. Thermostat

B. Resistance Temperature Detectors (RTDs)

C. Thermistors

D. Thermocouples

3. IR Thermal Radiation Sensors

4. Digital Temperature Sensor

<https://wiki.metropolia.fi/display/sensor/Temperature+Sensor>

1. Introduction

A temperature sensor is a device that measures the “hotness” (temperature) or heat content of a system. Sometimes they are also called thermal sensors.

Why do we care?

- Comfort -- at home and offices – minimal req'ment
- Weather / climate (might need to track T vs t) – robust and steady
- Health (body temperatures) – accurate and non-contact
- Industry application (e.g. food production, processing materials) – wide T range ...
- Science (might need to measure extreme temperatures) – most demanding: accuracy, sensitivity, T range...



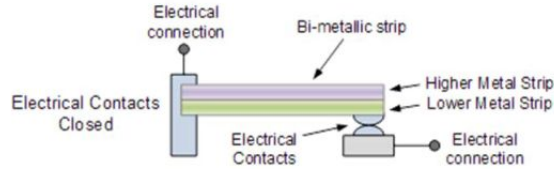
Source:

<https://sps.honeywell.com/gb/en/products/advanced-sensing-technologies/industrial-sensing/industrial-sensors/temperature-sensors>

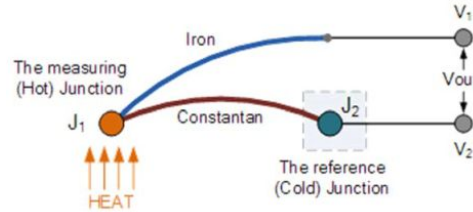
Temperature measuring devices

We'll discuss temperature sensors that can interact with electrical circuits (traditional mercury/alcohol thermometers are therefore excluded)

Thermostat (thermal electrical switch)



Thermocouple (min-battery)



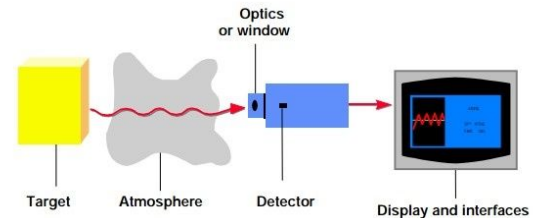
Thermistor



Resistive Temperature Detectors (RTD)

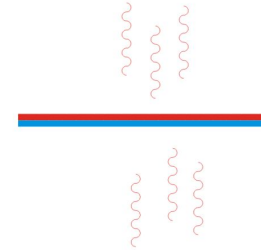
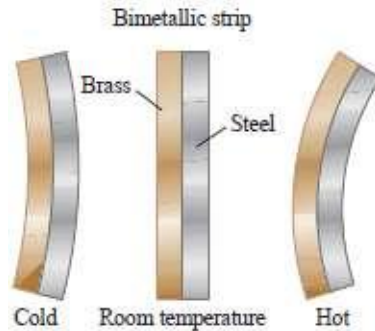


IR Sensors



2A. Thermostat

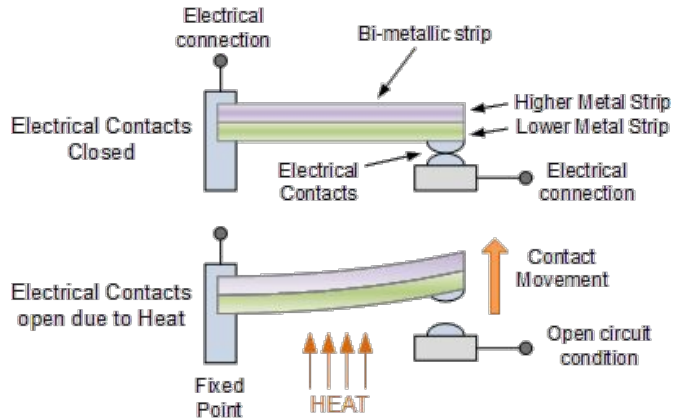
- A thermostat is a contact type electro-mechanical temperature switch.
- Consists of two different metals (e.g. Ni, Cu, W, Al) that are bonded together to form a bi-metallic strip.
- The different linear expansion rates of the two dissimilar metals produces a mechanical bending movement when the strip is subjected to heat.



<https://commons.wikimedia.org/wiki/File:Bimetal.gif>

Using a thermostat in a circuit

- When a thermostat is cold the contacts are closed. Current passes through the thermostat.
- When it is hot, one metal expands more than the other and the bonded bi-metallic strip bends up (or down) opening the contacts preventing the current from flowing.



<https://sps.honeywell.com/gb/en/products/advanced-sensing-technologies/industrial-sensing/industrial-sensors/temperature-sensors/thermostats>

2A. Thermostat applications

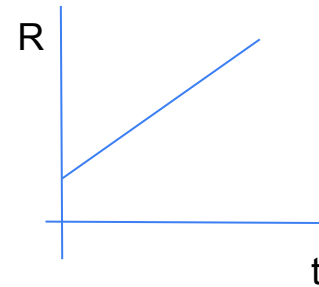
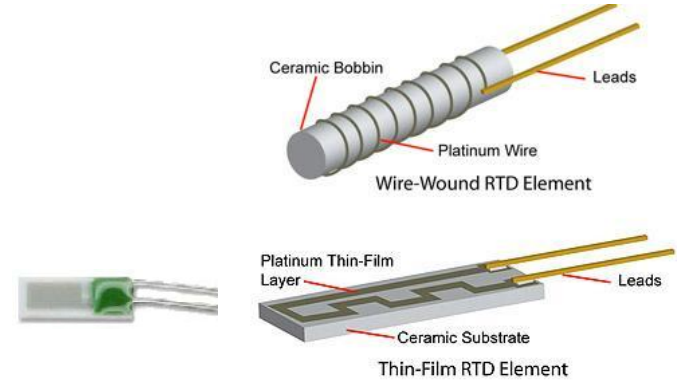
Thermostats are commonly used in offices / homes for controlling the temperature set point of ovens, irons, immersion hot water tanks and office A/C system.

They are cheap and are available over a wide operating range. But one main disadvantage is that they have a large hysteresis range from when the electrical contacts open until when they close again. For example, it may be set to 20°C but may not open until 22°C or close again until 18°C.



2B. Resistance **Temperature** Detector (RTD)

- RTD is a temperature dependent resistor made from high-purity conducting metals such as platinum, copper or nickel.
- They are wound into a coil and whose electrical resistance changes as a function of temperature.
- Also available are thin-film RTD's. These devices have a thin film of platinum paste deposited onto a white ceramic substrate.
- RTD have positive temperature coefficients (PTC), i.e. the resistance increases with increasing temperature. For some metals, e.g., Pt, the resistance value is extremely linear producing very accurate measurements of temperature. The operating temperature range is very wide: -200 to +600°C.



<https://www.designworldonline.com/designing-with-rtd-temperature-sensors/>

Why R increases with T?

Temperature dependent resistance values

Most metals follows approximately linear relationship above 0°C:

$$R = R_0(1 + \alpha t)$$

R_0 = resistance at 0°C

t = temperature in deg C

α = temperature coefficient [$^{\circ}\text{C}^{-1}$]

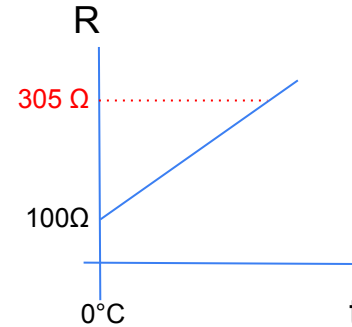
Typical α :

Pt: 0.0039 $^{\circ}\text{C}^{-1}$

Cu: 0.0043 $^{\circ}\text{C}^{-1}$

Ni: 0.0068 $^{\circ}\text{C}^{-1}$

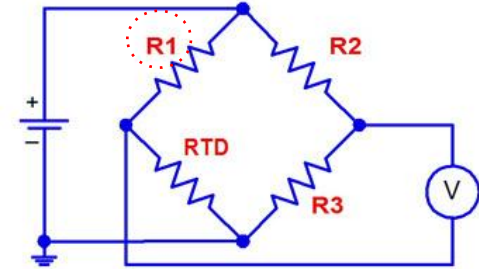
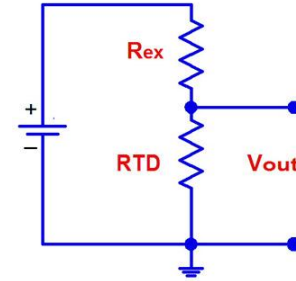
Means R changes by
0.39% per deg



A Pt thermometer has a resistance of 100Ω at 0°C. If the resistance becomes 305 Ω when it is in contact with a hot gas, determine the temperature of the gas. ($\alpha = 0.0039 ^{\circ}\text{C}^{-1}$)

How to use a RTD

- RTD's are passive resistive devices. To generate a voltage signal, wire up a RTD in a potential divider.
- A typical RTD has a resistance between 100-140 Ω for temperature range 0° -100°C
- But potential divider draws current I. There will be heating of the RTD because of I^2R heating of the resistor – i.e. the RTD becomes hotter, not because of true temperature change, but because the potential divider consumes electric power. We call this “self heating”.
- Self heating gives false signal. So, RTD is usually connected into a Wheatstone Bridge network to eliminate self heating. (Details of the Wheatstone bridge will be considered in Oct.)



RTD Advantages and Limitations

Advantages of RTD

- High accuracy
- Low drift
- Wide operating range: -200 to +600°C
- Best for precision applications.

Disadvantages:

- Cost.
- Self-Heating when high accuracy is needed
- Slow response

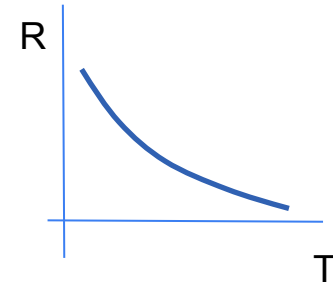


<https://sps.honeywell.com/gb/en/products/advanced-sensing-technologies/industrial-sensing/industrial-sensors/temperature-sensors>

2C. Thermistors

- A thermistor is a special type of resistor which respond to changes in temperature.
- Thermistors are generally made from **non-metallic materials**. (unlike RTD) Suitable materials are oxides of nickel, manganese or cobalt.
- The oxides behave as semiconductors. They are powders which can be pressed into discs or balls and then sealed.

Most thermistors have a Negative Temperature Coefficient of resistance or (NTC). Their resistance value goes DOWN with an increase in the temperature.



Why R decreases with T?

Typ. resistance values: from a few Ω to $M\Omega$. Most are $\sim k\Omega$.

Temperature dependent resistance values

Most thermistors follow an empirical temperature dependent R equation:

$$R = R_0 \exp\left[\beta\left(\frac{1}{T} - \frac{1}{T_0}\right)\right]$$

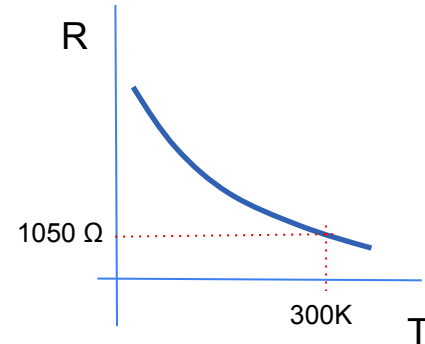
R_0 = resistance at an absolute temp T_0
 T = temperature in Kelvin
 β = temperature coefficient [K^{-1}]

Simple observation:

--- for a known T_0 and R_0 , if the temperature is raised to $T > T_0$, the exponent becomes negative, and $R < R_0$.

--- **So, R is reduced as T is increased, R will be increased as T is reduced.**

A thermistor has $\beta = 3140K^{-1}$, and the temperature at $27^\circ C$ is 1050Ω . It is then used to measure an unknown temperature T , and the measured resistance is 2330Ω . Find T .



How to use a thermistor

Thermistors are generally connected in series with another resistor to form a potential divider. The choice of resistor gives a voltage output at some pre-determined temperature point or value.

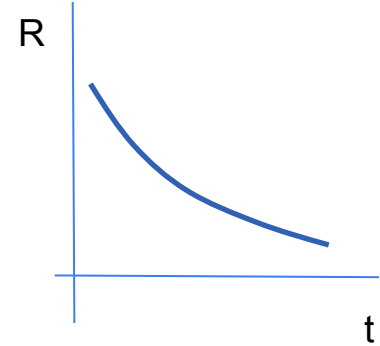
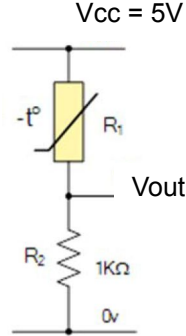
Example

A thermistor R_1 has a resistance value of $10\text{k}\Omega$ at 25°C and a value of 100Ω at 100°C . R_1 connected in a circuit as shown. Calculate V_{out} for both temperatures.

$$25^\circ\text{C} : V_{out} = \frac{1\text{k}}{10\text{k} + 1\text{k}} \times 5\text{V} = 0.454\text{V}$$

$$100^\circ\text{C} : V_{out} = \frac{1\text{k}}{0.1\text{k} + 1\text{k}} \times 5\text{V} = 4.545\text{V}$$

$$R = R_0 \exp\left[\beta\left(\frac{1}{T} - \frac{1}{T_0}\right)\right]$$



Assume the thermistor follows the thermistor eqn.

- Suppose the initial temperature is 25°C . What is T_0 ? What is R_0 ?
- What is β ?
- Find the potential divider output at 150°C .

Thermistor for high accuracy determination: the Steinhart model

Steinhart (1968) proposed a more accurate model equation for thermistor:

$$\frac{1}{T} = A + B \ln R_{th} + C \ln R_{th}^3$$

For 0-70°C, the **accuracy is within milli-degree** range. The model became an industry standard for calibrating precision thermistors.

To find coefficients (A, B, C) for the above equation, a system of three equations should be solved after a thermistor is calibrated at three temperatures, T_1 , T_2 , T_3 .

Practical thermistor specifications

Thermistors are available with lost cost with some standard specs.
Here, we look in to operation of **NTC-MF52** as an example.

$$R = R_0 \exp\left[\beta\left(\frac{1}{T} - \frac{1}{T_0}\right)\right]$$

Information extracted from specs:

- R25: 10 k Ω at 25°C.
- B is the β value. For R25 = 10 k Ω , β = 3950 K.
- So, T_0 (298K) and R_0 are specified when we buy the thermistor. If R is measured at an unknown T, T can be computed.

Normal specification Resistance & Temperature Table of MF52-type (Unit : K Ω)

	R25	10 K Ω	50 K Ω	100 K Ω	50 K Ω	50 K Ω	100 K Ω	100 K Ω	150 K Ω
T(°C) / Rt	B	3950	3950	4000	4050	4150	4150	4300	4500
-30		181.70	908.30	1790.00					
-25		133.30	666.50	1321.00					
-20		98.88	494.50	984.70					
-15		74.10	370.50	740.80					
-10		56.06	280.30	562.30					
-5		42.80	214.00	430.50					
0		98.96	164.80	332.30	168.80	172.00	344.10	352.40	576.70
5		25.58	127.90	257.50	131.30	132.20	264.30	270.00	433.20
10		20.00	99.98	201.10	101.00	102.40	204.80	208.30	328.40
15		15.76	78.79	158.20	79.28	80.03	160.10	161.90	250.90
20		12.51	62.55	125.40	62.78	63.00	125.00	136.70	193.30
25		10.00	50.00	100.00	50.00	50.00	100.00	100.00	150.00
30		8.048	40.24	80.29	39.98	39.76	79.51	78.35	117.30
35		6.518	32.59	64.87	32.16	31.89	63.77	62.37	92.28
40		5.312	26.56	57.72	26.10	25.73	51.45	49.94	73.11
45		4.354	21.77	43.10	21.35	20.88	41.76	40.22	58.28
50		3.588	17.94	35.42	17.72	17.04	34.08	32.56	46.74

Thermistor vs RTD

Adapted from: <https://www.omega.com/en-us/resources/rtd-vs-thermistors>

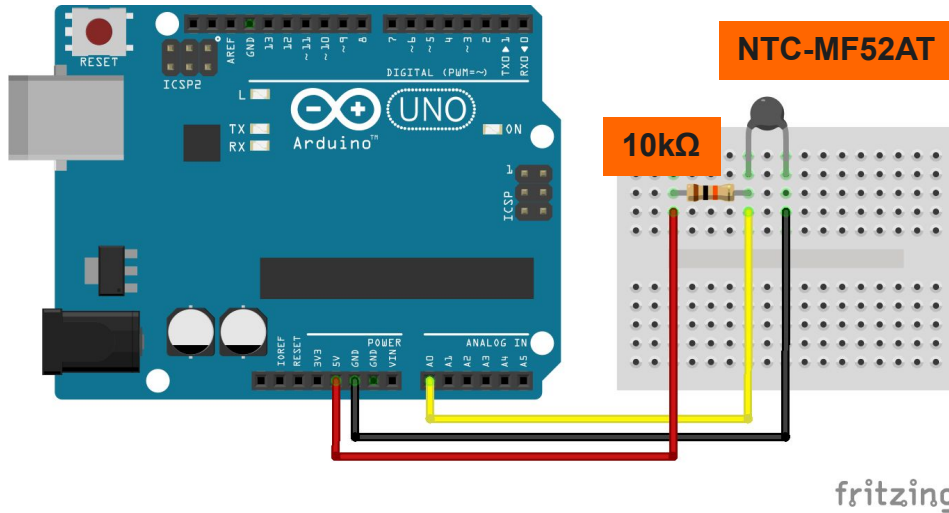
Sensor type	Thermistor	RTD
Temperature Range (typical)	-100 to 325°C	-200 to 650°C
Accuracy (typical)	0.05 to 1.5°C	0.1 to 1°C
Linearity	Exponential	Linear
Power required	Constant voltage or current	Constant voltage or current
Response time	Fast 0.12 to 10s	Generally slow 1 to 50s
Cost	Low to moderate	High

- Temperature range. If your application involves temperatures above 300°C, the RTD is your only option. Low temperatures: thermistors are often preferred.
- Thermistors are more accurate.

How to use thermistor with Arduino

Assemble the circuit:

a potential divider consisting of a thermistor, and a 10 k Ω resistor.



How to use thermistor with Arduino- software

****Complete the following code that is highlighted in yellow: ****

1. Input the known parameters from the thermistor specification e.g. β value, Resistance at certain temperature
2. The thermistor's temperature is computed from the temperature dependent R equation
3. Print the thermistor temperature in degree Celsius in the serial monitor (you may refer to Classwork 1B – Analog input)

```
//Thermometer with thermistor

//These values are in the datasheet
#define R_T1 ??? // Resistance from datasheet at temperature 25 degree Celsius
#define B ??? // Beta value from datasheet

#define VCC 5 //Supply voltage
#define R ??? //Resistor that is connected in series with thermistor

//Variables
float R_thermistor, V_R, In, T, Tc, V_thermistor;
float T1 = 25 + 273.15; //Temperature in Kelvin i.e 25 degree Celsius from datasheet
int sensor_value;

void setup() {
  Serial.begin(9600);
}
```

```
void loop()
{
  sensor_value = analogRead(A0); //Acquisition analog value of V_thermistor
  V_thermistor = (5.00/1023.00)*sensor_value; //voltage across thermistor
  V_R = VCC - V_thermistor; //voltage across R
  R_thermistor = V_thermistor/(V_R/R); //Resistance of thermistor

  T = ??? //Temperature from thermistor
  Tc = T - 273.15; //Conversion to Celsius
  ??? //Print the temperature in Celsius Tc in the serial monitor
  delay(500);
}
```

Temperature dependent R equation:

$$R = R_0 \exp\left[\beta\left(\frac{1}{T} - \frac{1}{T_0}\right)\right]$$

$$R_{T1} = (R_{\text{thermistor}}) \exp\left[B\left(\frac{1}{T1} - \frac{1}{T}\right)\right]$$

Class /homework: Thermometer by using a thermistor

Assemble the circuit without USB connection. The thermometer consists of two components

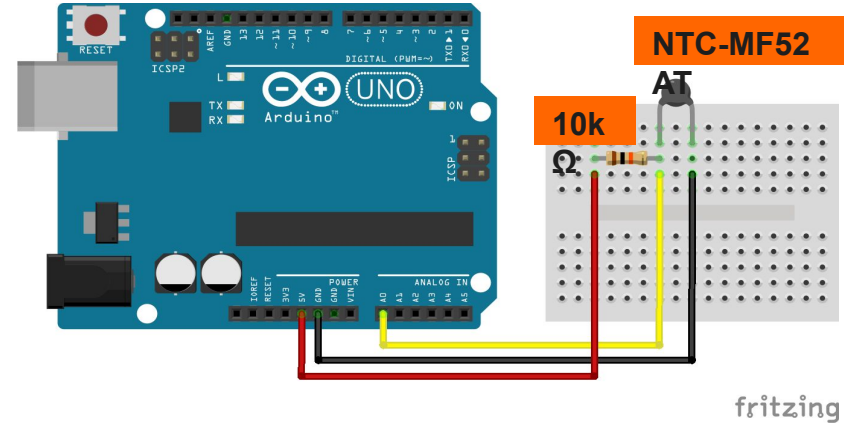
- (i) a NTC thermistor
- (ii) 10k Ω resistor

Your job:

1. Write a sketch and display the measured temperature from the thermistor at room temperature.
2. Immerse the thermistor into a cup of hot water. See how the measured temperatures changes.

Complete the assignment and submit:

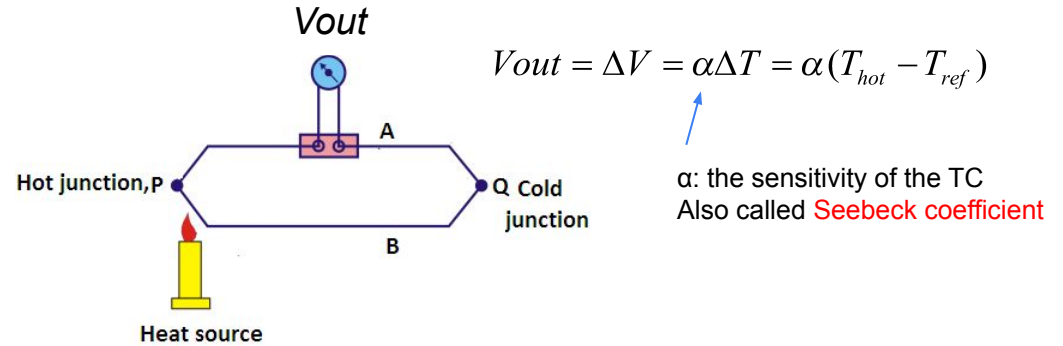
1. A photo of your circuit.
2. Your sketch
3. A temperature cooling plot of T vs Time
(at least 10 pairs of temperature data, time interval = 5 minutes)



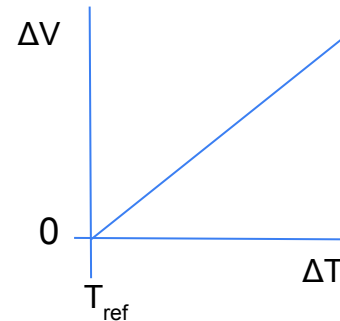
2D. Thermocouple

- Recall from Chap.2 the operating principle of a thermocouple (TC).

Two dissimilar metals 'A' and 'B' are joined at the two junctions 'P' and 'Q'. 'P' junction is measuring junction ("hot" junction), 'Q' is the reference junction or "cold" junction.

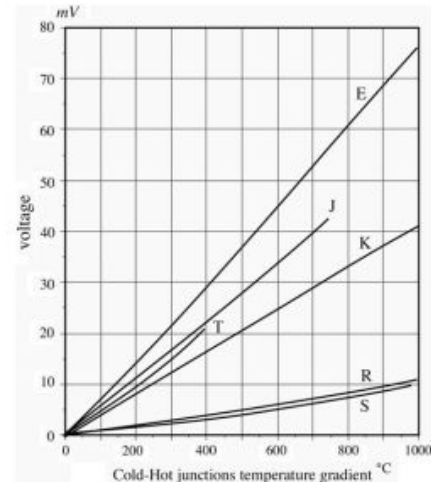
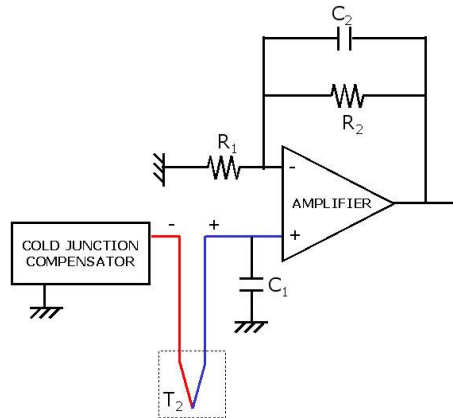
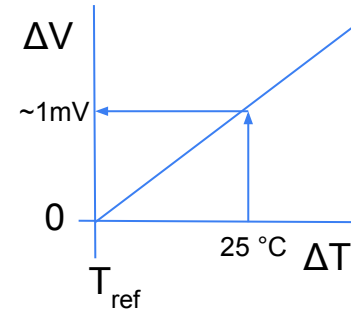


- Thermocouples are the most commonly-used thermal sensors in industrial, automobile and consumer electronics:
- They are inexpensive and have the widest temperature tolerance among all temperature sensors



Thermocouple signals need to be amplified

- The three most common TC materials are Iron-Constantan (Type J), Copper-Constantan (Type T), and Nickel-Chromium (Type K).
- The output voltage ΔV from a thermocouple is very small. E.g. K-type TC, at 25 °C, $\Delta V \approx 1\text{mV}$ if the reference junction is at °C. So each °C change will give about $1\text{mV}/25\text{ }^{\circ}\text{C} = 40\text{ }\mu\text{V}/^{\circ}\text{C}$.
- Output must be amplified for further processing.



Selected thermocouples and their characteristics

Thermocouple Sensor Colour Codes <i>Extension and Compensating Leads</i>			
Code Type	Conductors (+/-)	Sensitivity	British BS 1843:1952
E	Nickel Chromium / Constantan	-200 to 900°C	
J	Iron / Constantan	0 to 750°C	
K	Nickel Chromium / Nickel Aluminium	-200 to 1250°C	
N	Nicrosil / Nisil	0 to 1250°C	
T	Copper / Constantan	-200 to 350°C	
U	Copper / Copper Nickel	0 to 1450°C	

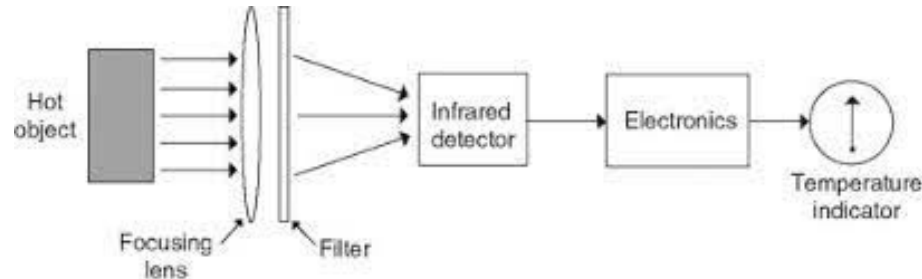
https://www.electronics-tutorials.ws/io/io_3.html

3. IR thermal radiation sensor

Temperature can be measured by contact and noncontact methods. Sensors in the last section mostly operate as contact or ambient sensors, i.e., the sensor is located right at the site.

However, there are needs to access environment which is difficult to get close to during a routine measurement. E.g. people with infectious diseases, hostile environments, unreachable terrestrial or non-terrestrial objects which are too dangerous for the operator, or just unreachable.

To measure temperatures remotely, we can measure thermal radiation from the object (blackbody radiation). Objects in daily life are between 0-40°C, and they emit in the IR range according to Wien's Law. Thus, thermometers built on sensing thermal radiation are sometimes call IR thermal sensors.



IR thermal radiation sensors

There are two main groups of IR thermal radiation sensors

Infrared Thermal Detectors

(incident ir radiation heats up the heat-sensitive detectors)

-- pyrometer

Thermal Quantum Detectors

(incident ir radiation causes changes in resistance, current etc)

-- IR photodiodes, IR phototransistors

- We'll focus here on the IR thermal detectors: **pyrometers**

Pyrometer (= remote heat sensing thermometer)

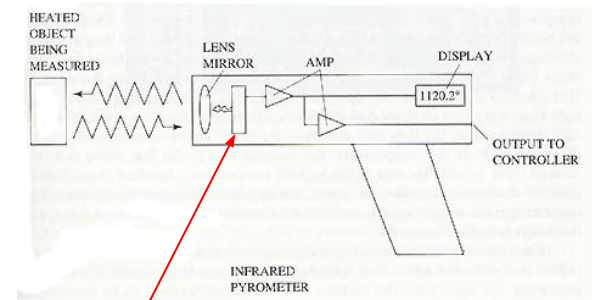
- Pyrometer can be considered to be a remote heat sensing thermometer.
- Pyrometer focuses thermal radiation from an object onto an IR sensor. The sensor output signal is related to the incident thermal radiation or irradiance (I), i.e., radiant energy per unit area.
- From I , the absolute temperature T of the object can be deduced through the Stefan–Boltzmann law:

$$I = \varepsilon \sigma T^4$$

T = temperature of object (in Kelvin)

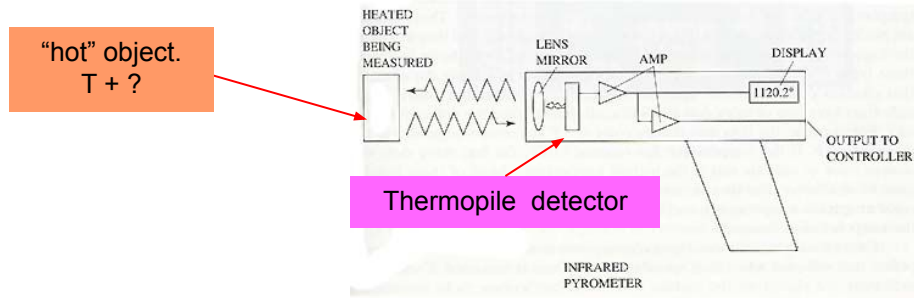
$\sigma = 5.67 \times 10^{-8} \text{ W} \cdot \text{m}^{-2} \cdot \text{K}^{-4}$ is the Stefan–Boltzmann constant

ε = is the emissivity of an object (ideal = 1 if it is a BB)

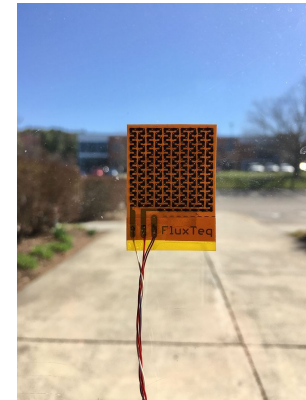
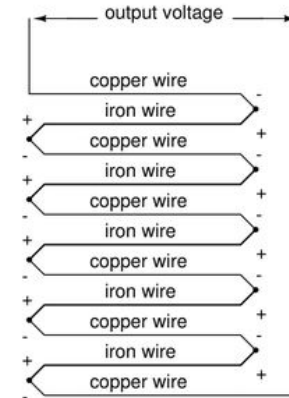


Detector turns incident IR into heat energy.

Thermopile detector for pyrometer



- Heat produced from incident IR radiation can be transformed into electrical signal using thermocouples (TC).
- But a single TC has small voltage output. The voltage can be increased a great deal by connecting many TC in series (called a thermopile).
- The output of the thermopile can be calibrated to indicate to indicate temperature of the IR source.

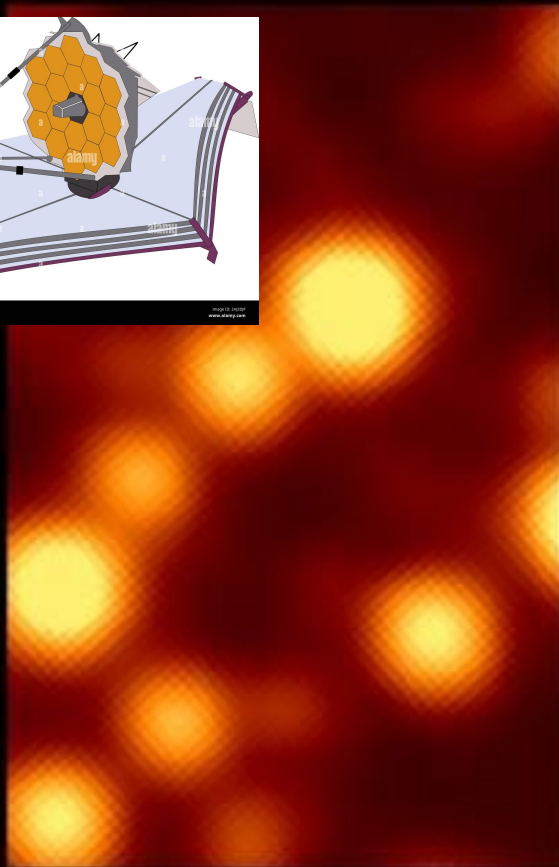
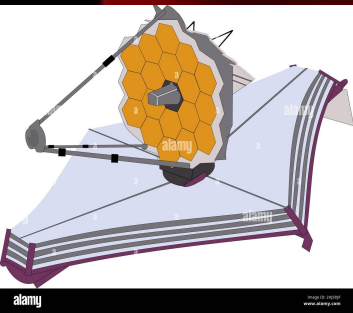


IR Imaging

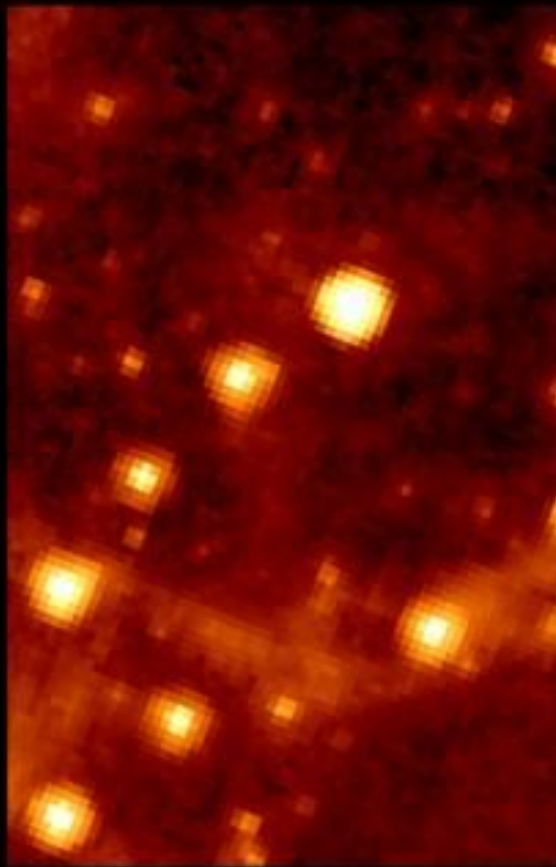
- CCD or CMOS image sensor for common camera are designed for visible lights.
- But they can be easily extended to the IR range. With IR sensing elements, IR image can be formed, and they have very important applications in everyday life, medical, scientific research, astronomy, and military applications.
- Use your imaginations!



The Evolution of Infrared Space Telescopes



WISE W2 4.6 μm



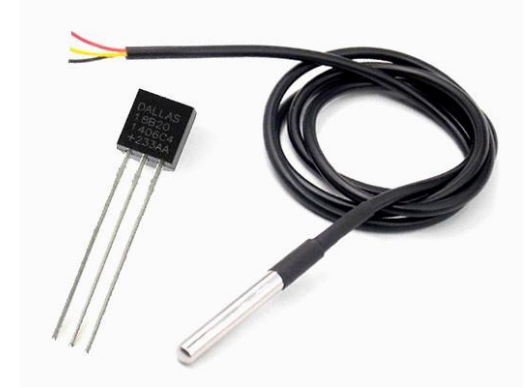
Spitzer/IRAC 8.6 μm



JWST/MIRI 7.7 μm

4. Digital Temperature Sensor

- A modern trend in sensor development is digital sensors, whose output is intrinsically in the form of numbers rather than voltage or current.
- These digital sensors have their own built-in microcontrollers and conversion circuitry, which convert sensor signals into numbers – a **“smart sensor”**. Later on in the course, we’ll build our own smart sensor.
- You can use Arduino to talk to digital sensors directly. So no need to use build additional circuitry (e.g. potential dividers); no need to use analogRead, related conversion statements) have been placed on the sensor itself.
- Here, we look into **digital temperature sensor DS18B20** device. It consists of an internal temperature dependent resistor, a small processor for converting the signal into digital form, and memory to store data. After the built-in sensor on makes a temperature measurement, it communicates the result digitally, using the 1-wire protocol.



4. How to use the DS18B20 temperature sensor

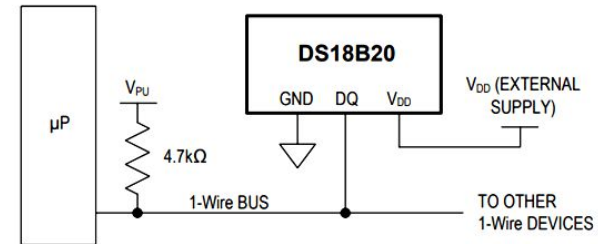
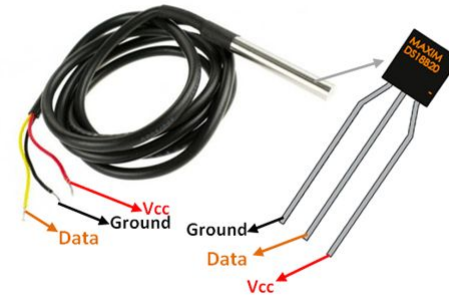
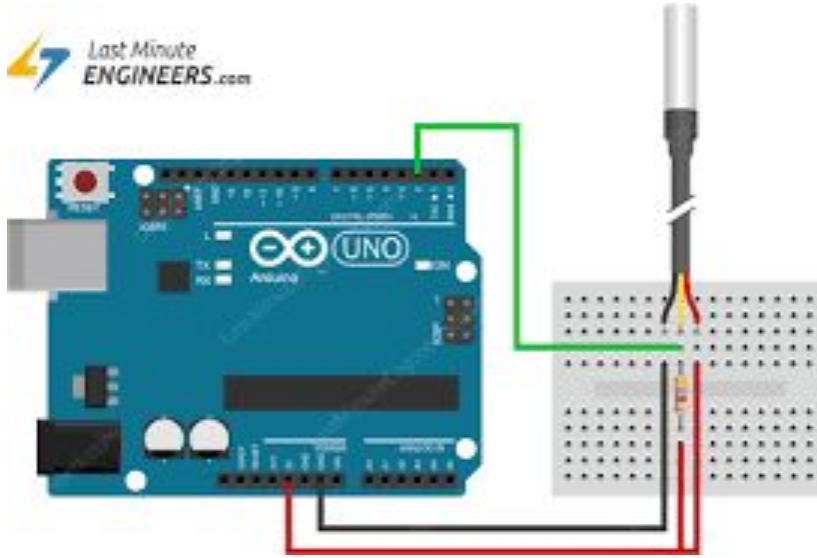
DS18B20 Sensor Specifications

- Communicates using 1-Wire method
- Operating voltage: 3V to 5V
- Temperature Range: -55°C to $+125^{\circ}\text{C}$
- Accuracy: $\pm 0.5^{\circ}\text{C}$
- Output Resolution: 9-bit to 12-bit (programmable)
- Can multiplex
- Conversion time: 750ms at 12-bit

Available as To-92, SOP and even as a waterproof sensor



4. How to use the DS18B20 temperature sensor – the hardware



4. How to use the DS18B20 temperature sensor – software

To run the codes below, go to the link at the bottom of the page and follow instructions to install 2 libraries: <OneWire.h> and <DallasTemperature.h>

```
// One-wire T sensor DS18B20
#include <OneWire.h>
#include <DallasTemperature.h>

// Data wire is plugged into digital pin 2 on the Arduino
#define ONE_WIRE_BUS 2

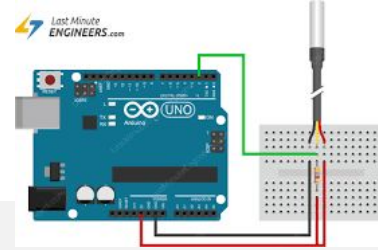
// Setup a oneWire instance to communicate with any
// OneWire device
OneWire oneWire(ONE_WIRE_BUS);

// Pass oneWire reference to DallasTemperature library
DallasTemperature sensors(&oneWire);

void setup(void)
{
  sensors.begin(); // Start up the library
  Serial.begin(9600);
}
```

```
void loop(void)
{
  // Send the command to get temperatures
  sensors.requestTemperatures();

  //print the temperature in Celsius
  Serial.print("Temperature: ");
  Serial.print(sensors.getTempCByIndex(0));
  Serial.print((char)176); //shows degrees character
  Serial.print("C | ");
  //print the temperature in Fahrenheit
  Serial.print((sensors.getTempCByIndex(0) * 9.0) / 5.0 + 32.0);
  Serial.print((char)176); //shows degrees character
  Serial.println("F");
  delay(500);
}
```



Class work: Digital temperature sensor

Obtain a DS18B20 from your instructor. Assemble the circuit. Key in the program in Arduino. Make sure you install 2 libraries `<OneWire.h>` and `<DallasTemperature.h>`.

Your job:

1. Compile the sketch and observe the displayed temperature as you warm the temperature probe with you hand.
2. Hold your probe in hand. Open the serial plotter. Watch the temperature rise to a steady value. Take out the probe and observe the falling temperatures as the probe cools.

